

Travor Zihao Liu
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Stanford, California, USA

Education

- 2025– **Ph.D.**, Mathematics, Stanford University
- 2022–2025 **B.Sc.** Mathematics, University College London
- First Class, Ranked 1st out of 341

Research Experience

- 2025– Value distributions of L -functions (Supervisor: [Dr. Kannan Soundararajan](#))
- Investigating topics in the distribution of values of L -functions in the critical strip and applications to number theory.
- 2025– Higher Order Fourier Analysis (Supervisor: [Dr. Sarah Peluse](#))
- Exploring Gowers norm and its applications to the theorems of Szemerédi, Green–Tao, and Green–Sawhney
- 2024–2025 Explicit Quadratic Large Sieve Inequality, (Supervisor: [Dr. Ian Petrow](#))
- Funded by LMS Undergraduate Research Bursaries (URB-2024-41)
 - Followed the work of Heath-Brown and produced a research paper
- 2024 Special Values of the Riemann Zeta Function (Supervisor: [Dr. Cecilia Busuioc](#))
- Investigated Bernoulli numbers and computed $\zeta(2k)$ for integers $k \geq 1$
 - Studied Beukers’s proof that $\zeta(3)$ is irrational
- 2023–2024 p -adic Numbers (Supervisor: [Dr. Cecilia Busuioc](#))
- With Hantang Guo, May Jiang, and Qing Su
 - Proved Ostrowski’s theorem and p -adic Weierstrass preparation theorem
 - Wrote Python code for visualization
- 2023 Axiom of Choice (Supervisor: [Dr. Cecilia Busuioc](#))

- With Yi Liu, Daya Singh, and Tairan Wang
 - Investigated equivalent forms, consequences, and the independence of the Axiom of Choice from ZF set theory
 - Studied well-ordered sets, ordinals, and transfinite recursion
- 2021 Prime Number Theorem
- Developed a new method to prove the prime number theorem
 - Published an academic research paper and presented at ICPAM 2021
- 2020 PM2.5 Density Prediction
- Studied PM2.5 density using signal processing, Pandas, and LightGBM
 - Innovated and published a paper on a new PM2.5 prediction model
- 2017–2019 [Operating System Development](#)
- Created two 32-bit operating systems from scratch
 - Implemented keyboard driver and virtual memory using C and x86 assembly

Publications

1. Liu, Z. *Explicit quadratic large sieve inequality* Accepted by Acta Arithmetica. May 2025. <http://arxiv.org/abs/2505.09637>.
2. Liu, Z. *On a weighted sum over multiplicative functions and its applications to the GPY sieve* arXiv:2210.01671 [math]. Mar. 2023. <http://arxiv.org/abs/2210.01671>.
3. Liu, Z. A Direct Proof of the Prime Number Theorem using Riemann's Prime-counting Function. *Journal of Physics: Conference Series* **2287**. Publisher: IOP Publishing, 012008. <https://doi.org/10.1088/1742-6596/2287/1/012008> (2022).
4. Liu, Z. *A Corrected Simplified Proof of Chen's Theorem* arXiv:2203.07871 [math]. Mar. 2022. <http://arxiv.org/abs/2203.07871>.
5. Liu, Z. *PM2.5 Density Prediction Based on a Two-Stage Rolling Forecast Model Using LightGBM in Computing and Data Science* (eds Cao, W., Ozcan, A., Xie, H. & Guan, B.) (Springer Nature Singapore, Singapore, 2021), 228–248. ISBN: 978-981-16-8885-0.

Presentations

Student-Run Seminars on Number Theory and Function Theory

1. [On the distribution of zeros of \$\zeta\(s\)\$](#) . July 2025.
2. Ω results in number theory. July 2023.
3. GPY sieve - from Selberg to Maynard. July 2022.

UCL Undergraduate Math Colloquium

1. *Special values of the Riemann zeta function*. Oct. 2024.

2. *Introduction to p -adic analysis*. Feb. 2024.
3. *Ray reflection and rational approximation*. Oct. 2023.
4. *Ellipses, pendulum, and double periodicity*. Mar. 2023.
5. *Bounded gaps between primes*. Oct. 2022.

UCL Undergraduate Project Talks

1. *Inverse Galois Problem*. Dec. 2024.
2. *Axiom of Choice: Equivalents, Consequences, and Independence*. June 2023.

Teaching

Stanford University

Teaching Assistantships

2025–2026 Math 104 Applied Matrix Theory (Autumn, Spring)

Students Supervised

2026	Karthik Seetharaman	Directed Reading Program
2026	Robert Dragomirescu	Directed Reading Program

Attended Seminars

2025	Summer School on Analytic Number Theory (Xi'an, China)
2024	International Number Theory Conference in Commemoration of Chengdong Pan (Jinan, China)
2022–2025	London Number Theory Seminars
2022–2025	London Junior Number Theory Seminars
2022–2025	London Heilbronn Colloquia

Other Experience

2020–	Math Stack Exchange and Math Overflow • Answered 250+ questions on complex analysis and number theory • Earned 7.5k+ score MSE and 1.3k+ scores on MO as of 2025 • Top answerer under the tag “analytic number theory”
2020–	Math Blogging on Zhihu

- Composed 160+ articles for 8 self-initiated math columns
- Answered 980+ math questions from netizens
- Obtained 82k+ followers, 87k+ upvotes, and 77k bookmarks as of 2025

Awards & Honors

2025–2026	IMA Membership
2025	UCL Dean’s List Nomination
2025	Wynne Roberts Prize (Best graduating math student)
2024	LMS Undergraduate Research Bursaries (URB-2024-41)
2024	MSci Prize in Mathematics (Best Year 2 overall performance)
2024	Kestelman Prize (Best Year 2 performance in Real and Complex Analysis)
2023	Filon Prize (Best Year 1 overall performance)
2022	Platinum Division, USA Computing Olympiad
2021	Bronze, S.-T. Yau High School Science Award (Mathematics)
2019	Finalist Award, High School Mathematical Contest in Modeling

Last updated: April 9, 2026